Fitting a curve to the dam — Results

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Chosen model: **logistic** — h(t) = c + a / (1 + exp(-k (t - t0)))

Fitted parameters

Parameter	Value	Std. Error	t statistic	p-value
c	14.3775	0.0379	379.308	0
a	5.5139	0.0499	110.407	0
k	0.8555	0.0625	13.695	0
t0	30.1577	0.0978	308.243	0

Fit statistics

n	p	dof	SSE (m2)	SST (m2)	R2	s (m)
288	4	284	42.41254	2034.07794	0.97915	0.3864

Area under the fitted level

Quad integral A	Abs. error (quad)	Trapezoid cross-check	Difference
1260.92388 m·h	4.57e-07 m·h	1260.99625 m·h	-0.07237 m·h

Units: h in meters, t in hours → area is in meter-hours, not a volume.

Reading of the residuals

Centred / drift: TODO: is the scatter centred on zero throughout, or does it drift?
Runs: TODO: are there long runs of the same sign? What does that say about the model shape?
Spread: TODO: does the spread widen as the level rises?
Vs. logger resolution (1 cm): TODO: is the largest residual bigger than the logger's 1 cm resolution, and by how much?